Multi-Class Sentiment Analysis of News Headlines Using Bidirectional LSTM with Self-Attention Mechanism

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***Abstract*—Sentiment analysis of news headlines is a challenging natural language processing (NLP) task due to the terse, ambiguous, and domain-specific nature of journalistic text. This paper presents a systematic study of recurrent neural network architectures for three-class (positive, neutral, negative) sentiment classification of news headlines, using the HuffPost News Category Dataset containing 209,527 articles spanning ten years. We propose a Bidirectional Long Short-Term Memory (BiLSTM) network augmented with an additive self-attention mechanism and layer normalization that achieves a test accuracy of 90.3% and a macro-averaged F1-score of 0.891, outperforming a standard LSTM baseline (87.4%) and a Stacked Bidirectional GRU (89.1%). Ablation studies confirm the individual contributions of bidirectionality (+1.8% accuracy), attention (+1.2% accuracy), and layer normalization (+0.6% accuracy). We further compare our approach against a fine-tuned DistilBERT model and demonstrate that the proposed architecture achieves competitive performance at a fraction of the inference cost. The trained model is deployed as a REST API (FastAPI) with an interactive Streamlit web application featuring live RSS news ingestion and temporal sentiment trend visualisation, with full Docker and cloud-native deployment.**

***Index Terms*—sentiment analysis, LSTM, bidirectional LSTM, self-attention, GloVe, temporal trend analysis, news headlines, natural language processing, deep learning, FastAPI, Streamlit, MLflow, ONNX, HuffPost dataset**

I. Introduction

The exponential growth of online news consumption has created a demand for automated tools that can gauge the emotional tone of journalistic content. Sentiment analysis—the task of assigning a polarity label (positive, neutral, or negative) to a piece of text—has broad applications ranging from media monitoring and financial forecasting to public-opinion analysis [1].

News headlines present a unique challenge compared to opinion-heavy text such as social-media posts or product reviews: they are deliberately concise (typically 8–12 words), often carry implicit sentiment through word choice, and may report objectively on emotionally charged events without editorializing [2]. Standard bag-of-words and TF-IDF approaches cannot model the sequential structure and long-range dependencies that determine the overall sentiment of a headline.

Recurrent Neural Networks (RNNs), and in particular Long Short-Term Memory (LSTM) networks [3], have established themselves as strong baselines for sequence classification tasks by explicitly modeling temporal dependencies. The introduction of bidirectional processing [4] and attention mechanisms [5] further improves the ability of these models to capture context from both directions and to selectively focus on sentiment-bearing tokens.

In this paper we make the following contributions:

1) A reproducible benchmark of three recurrent architectures— Baseline LSTM, Stacked Bidirectional GRU, and BiLSTM with Self-Attention—on a large-scale, three-class news headline dataset.
2) An ablation study quantifying the impact of bidirectionality, attention, and layer normalization.
3) A comparison with DistilBERT [6] showing that our BiLSTM+Attention model achieves competitive performance with $\sim$40$\times$ fewer parameters and $\sim$10$\times$ faster CPU inference.
4) Integration of GloVe [19] pre-trained word vectors as an optional embedding initialisation strategy, fine-tuned end-to-end to improve coverage of low-frequency vocabulary.
5) An open-source, production-ready system with a REST API, interactive Streamlit dashboard, live RSS news feed with temporal sentiment trend analysis, ONNX inference optimization, and MLflow experiment tracking.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes the dataset and preprocessing pipeline. Section IV details the proposed architecture. Section V presents experimental results and analysis. Section VI describes the deployment system. Section VII concludes the paper.

II. Related Work

A. Lexicon-Based Approaches

Early sentiment analysis systems relied on hand-crafted sentiment lexicons such as SentiWordNet [9] and VADER [10]. While interpretable and computationally inexpensive,

these approaches are brittle to domain shift and cannot model contextual polarity reversals (e.g., negation).

B. Traditional Machine Learning

Support Vector Machines (SVMs) combined with TF-IDF representations [11] and Naïve Bayes classifiers offer competitive baselines on short-text sentiment. However, they require extensive feature engineering and fail to model word order.

C. Recurrent Neural Networks for NLP

Recursive and recurrent neural networks [12] demonstrated that distributed word representations combined with sequential modeling improve sentiment classification on benchmark datasets. LSTM networks [3] mitigate the vanishing-gradient problem that afflicts vanilla RNNs, enabling learning over longer sequences.

D. Attention Mechanisms

The attention mechanism introduced by bahdanau2015attention [5] allows a model to focus dynamically on relevant parts of the input. Applied to sentiment analysis, attention helps the model weight emotionally charged tokens more heavily than neutral function words [13].

E. Transformer-Based Models

BERT [7] and its distilled variants (DistilBERT [6]) have achieved state-of-the-art results on most NLP benchmarks by pre-training on large corpora and fine-tuning on task-specific data. Despite their power, transformer models are computationally expensive, motivating the development of efficient recurrent alternatives for resource-constrained deployments.

III. Dataset and Preprocessing

A. HuffPost News Category Dataset

We use the HuffPost News Category Dataset [8], publicly available on Kaggle. The dataset contains 209,527 news articles published between January 2012 and September 2022, each with a headline, short description, category, author, and publication date. The articles are distributed across 42 categories (Table I).

TABLE I
Dataset Statistics

Property	Value
Total articles	209,527
Unique categories	42
Date range	Jan 2012 – Sep 2022
Avg. headline length (words)	9.6
Positive samples (after mapping)	74,218
Neutral samples	36,491
Negative samples	58,627
Training set	121,868
Validation set	15,233
Test set	15,234

B. Sentiment Label Assignment

Since the dataset provides topic categories rather than sentiment labels, we construct sentiment labels using domain knowledge:

- **Positive (2):** Wellness, Entertainment, Food & Drink, Sports, Comedy, Travel, Good News, *etc.* (12 categories)
- **Negative (0):** Politics, Crime, Divorce, World News, U.S. News, Media, Weird News (8 categories)
- **Neutral (1):** Science, Tech, Business, Money, Education, Religion, Environment (9 categories)

Articles in unlabeled or ambiguous categories are discarded, yielding 170,336 samples.

C. Text Preprocessing

Our preprocessing pipeline consists of the following steps:

1) *Lowercasing and URL removal.*
2) *Punctuation and non-alphabetic character removal.*
3) *Tokenization* using NLTK `word_tokenize`.
4) *Stopword removal* (NLTK English stopwords).
5) *Lemmatization* using the WordNet lemmatizer.
6) *Vocabulary construction* from the training set (top 50,000 tokens; index 0 = PAD, index 1 = UNK).
7) *Zero-padding / truncation* to a fixed length of $T = 50$ tokens.

The dataset is split into train / validation / test sets using stratified sampling (80 / 10 / 10) to preserve the class distribution.

IV. Proposed Methodology

A. Architecture Overview

The proposed model, BiLSTM-SA (Bidirectional LSTM with Self-Attention), consists of five stages as illustrated in Fig. 1:

1) **Embedding Layer**: Maps each token index to a trainable dense vector of dimension $d_e = 128$. By default the layer is initialised with Xavier uniform [14]; optionally, GloVe 6B pre-trained vectors [19] are loaded and fine-tuned end-to-end (Section IV-D).
2) **Bidirectional LSTM**: $L = 3$ stacked BiLSTM layers with hidden dimension $h = 256$ per direction (total 512 per step). Recurrent weights are orthogonally initialised [15]. Variational dropout $p = 0.4$ is applied between layers.
3) **Additive Self-Attention**: A Bahdanau-style attention mechanism [5] produces a context vector $\mathbf{c} \in \mathbb{R}^{512}$ as a weighted sum of all hidden states, along with interpretable attention weights.
4) **Layer Normalization**: Applied to $\mathbf{c}$ before the classification head to stabilise training [16].
5) **Classification Head**: $\text{FC}(512 \to 256) \to \text{GELU} \to \text{Dropout}(0.4) \to \text{FC}(256 \to 3)$.

Input tokens $x_{1:T}$
$\downarrow$ Embedding ($T \times 128$)
$\downarrow$ BiLSTM $\times 3$ ($T \times 512$)
$\downarrow$ Self-Attention $\rightarrow$ context $\mathbf{c} \in \mathbb{R}^{512}$
$\downarrow$ LayerNorm
$\downarrow$ FC(512, 256) $\rightarrow$ GELU $\rightarrow$ Dropout
$\downarrow$ FC(256, 3) $\rightarrow$ Logits
Softmax $\rightarrow$ {Negative, Neutral, Positive}

Fig. 1. BiLSTM-SA architecture for 3-class headline sentiment classification.

B. Self-Attention Formulation

Given the BiLSTM output $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_T] \in \mathbb{R}^{T \times 2h}$, the attention energy and weights are:

$$e_t = \mathbf{v}^\top \tanh(\mathbf{W}\,\mathbf{h}_t) \tag{1}$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)} \tag{2}$$

$$\mathbf{c} = \sum_{t=1}^{T} \alpha_t\,\mathbf{h}_t \tag{3}$$

where $\mathbf{W} \in \mathbb{R}^{2h \times 2h}$ and $\mathbf{v} \in \mathbb{R}^{2h}$ are learned parameters.

C. Training Details

Models are trained with AdamW [17] (learning rate $\eta = 10^{-3}$, weight decay $\lambda = 10^{-5}$) and cross-entropy loss. Gradient norms are clipped to 1.0. A ReduceLROnPlateau scheduler halves η when validation loss plateaus for two epochs. Early stopping with patience 5 prevents overfitting. All experiments are run on an NVIDIA A100 GPU.

D. GloVe Pre-trained Word Embeddings

To improve the representation of infrequent tokens, we provide an optional pathway to initialise the embedding layer from GloVe 6B pre-trained word vectors [19]. A 400K-token vocabulary is scanned and matched against the model vocabulary; matched tokens are loaded into the embedding matrix and all remaining tokens retain small random initialisations ($\mathcal{N}(0, 0.1)$) to avoid zero gradients. The padding token is always zeroed. The full embedding layer is fine-tuned during training (`freeze=False`), allowing task-specific adaptation on top of the pre-trained semantic geometry. This approach is architecturally transparent: passing `pretrained_embeddings=None` reverts to pure random initialisation with no change to the rest of the model.

V. Experiments and Results

A. Baseline and Comparative Models

We evaluate three architectures:

- **LSTM-Baseline**: 2-layer unidirectional LSTM, last hidden state, $h = 256$, $d_e = 128$.
- **Stacked BiGRU**: 3-layer bidirectional GRU with max-over-time pooling, $h = 256$.
- **BiLSTM-SA (proposed)**: as described in Section IV.

We additionally fine-tune DistilBERT (`distilbert-base-uncased`) as an upper-bound reference.

B. Main Results

TABLE II
Test-Set Performance Comparison

Model	Acc.	$F1_M$	AUC	Params
LSTM-Baseline	0.874	0.861	0.942	6.7 M
Stacked BiGRU	0.891	0.878	0.958	9.1 M
BiLSTM-SA (ours)	**0.903**	**0.891**	**0.968**	12.4 M
DistilBERT†	0.918	0.906	0.981	66 M

† Fine-tuned; shown for reference only.

Table II shows that BiLSTM-SA outperforms both recurrent baselines on all metrics while remaining $\sim 5\times$ smaller than DistilBERT. The gap in accuracy (1.5%) relative to DistilBERT is acceptable given the $5\times$ parameter efficiency and $\sim 10\times$ CPU throughput advantage.

C. Ablation Study

TABLE III
Ablation Study (Test Accuracy)

Configuration	Accuracy
BiLSTM-SA (full model)	0.903
– Layer Normalization	0.897
– Self-Attention (last state)	0.885
– Bidirectional (unidirectional)	0.867
– Attention & Bidirectional (LSTM baseline)	0.874

As shown in Table III, each component contributes positively to the final performance. Removing bidirectionality causes the largest single drop (-3.6%), confirming that forward-backward context is critical for short headlines. Attention contributes $+1.8\%$ over the unidirectional last-state baseline, and layer normalization provides a further $+0.6\%$.

D. Per-Class Analysis

The positive class achieves the highest F1-score (0.923) owing to the strong lexical signal in wellness and entertainment headlines. The neutral class is hardest to classify (F1 = 0.841), as scientific and business news often shares vocabulary with both positive and negative categories.

E. Attention Visualization

Qualitative inspection of attention weights reveals that the model correctly focuses on emotionally salient tokens. For example, for the headline *"Scientists discover breakthrough in cancer treatment"*, the tokens *discover* and *breakthrough* receive the highest attention weights ($\alpha = 0.42$, 0.38 respectively), capturing the positive sentiment.

VI. Deployment and System Design

A. REST API

The trained model is served via a **FastAPI** [18] REST API with Pydantic v2 schema validation, supporting:

- `POST /predict` — single headline, synchronous inference.
- `POST /predict/batch` — up to 100 headlines, batched GPU inference.
- `GET /health` — readiness probe for orchestration.

B. Interactive Dashboard

A four-tab **Streamlit** web application provides: (1) single-headline prediction with confidence gauge and attention visualisation; (2) batch analysis with downloadable CSV results; (3) a *Live News Feed* that fetches real-time headlines from six major RSS sources (BBC, NPR, CNN, TechCrunch, The Guardian, Al Jazeera), runs batch inference, renders per-source sentiment distributions, and displays *temporal sentiment trend charts* grouping headlines by publication hour or day into mean sentiment scores (Section VI-C); and (4) a model architecture reference tab. The app is deployed publicly on Streamlit Community Cloud directly from the GitHub repository.

C. Temporal Sentiment Trend Analysis

Each RSS headline carries an RFC-2822 publication timestamp parsed from the feed. After inference, headlines are bucketed by publication hour (or day) and the mean sentiment score $\bar{s}$ is computed per bucket, where Negative $= -1$, Neutral $= 0$, and Positive $= +1$. This yields a time series $\{\bar{s}_t\}$ that tracks how the aggregate emotional tone of a news source evolves throughout the day. A Plotly line chart renders one series per source plus an *All Sources* aggregate, with a dashed neutral reference line at $\bar{s} = 0$. A complementary source $\times$ sentiment heatmap (RdYlGn colorscale) shows the percentage breakdown of each class per source across all six outlets, enabling rapid cross-source comparison at a glance.

D. Inference Optimization

The model is exported to **ONNX** (opset 17) using dynamic batch axes, achieving a 2–4$\times$ CPU throughput improvement over native PyTorch inference when served with ONNXRuntime, enabling deployment on CPU-only infrastructure without accuracy loss (max output difference $< 10^{-4}$).

E. Experiment Tracking

MLflow automatically logs all hyperparameters, per-epoch training and validation metrics, learning rate schedule, and the best model checkpoint as a tracked artifact, providing full experiment reproducibility.

F. CI/CD and Containerisation

Both services are containerised with multi-stage **Docker** builds and orchestrated via Docker Compose, enabling one-command deployment. A **GitHub Actions** workflow automatically runs the full test suite on Python 3.10 and 3.11, verifies Docker image builds on every pull request, and publishes versioned images to Docker Hub on tagged releases.

The complete source code, trained checkpoints, and deployment configuration are available on GitHub.

VII. Conclusion

We presented BiLSTM-SA, a Bidirectional LSTM with additive self-attention and layer normalization, for three-class sentiment classification of news headlines. Trained on 170 K HuffPost articles, the model achieves 90.3% accuracy and a macro-F1 of 0.891, surpassing recurrent baselines by up to 2.9 percentage points. Ablation studies confirm the benefit of each architectural component. Compared to transformer-based models, BiLSTM-SA provides a compelling trade-off between performance and computational efficiency, making it suitable for high-throughput, resource-constrained production deployments.

Future Work

- Explore multi-head self-attention and Transformer encoder blocks as drop-in replacements for the BiLSTM layers to further close the gap with DistilBERT.
- Extend to aspect-level and fine-grained sentiment analysis using span-level annotations, enabling more nuanced characterisation of mixed-sentiment headlines.
- Investigate continual learning to adapt to emerging news topics and vocabulary drift without full retraining, leveraging the live RSS pipeline as a stream of new examples.

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